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Mathematical Models

Maths is used to make sense of the world around us, and help us to make decisions, in our personal lives and in the worlds of business, politics, science, and so on. This could involve, for example:

- Creating and solving equations.
- Plotting graphs.
- Making calculations.
- Using computer software.

Each of these things could be thought of as creating a **mathematical model**. This chapter will introduce and explore a range of different kinds of model that may be useful, and when they may be encountered in the real world.

1.1 Systematic Estimation

It may be tempting to think of mathematics as a field devoted to calculating *exact*, *correct* answers.

Quote

"All models are wrong, but some are useful."
George Box, Statistician

The above quote highlights that mathematical models don't always need to be able to produce 'the *right* answer', and in many situations such a thing is not even possible. Instead, the goal is often a simply a *useful* answer.

The 20th century physicist Enrico Fermi was known for performing 'back of an envelope' calculations which often gave surprisingly reliable and *accurate* results. Calculations of this type are therefore sometimes called *'Fermi estimates'*. The goal of these *systematic estimates* is to quickly get a sense of the *scale* of numerical answers to complex or even impossible questions such as:

Questions

- *How many times might a teacher take a class register in their career?*
- *How much on average does a large supermarket spend on staff wages each week?*
- *How many miles are driven by cars on Scottish roads in a month?*

These calculations might involve using a mixture of *exact*, *approximated* and **assumed** values. For example:

- There are *exactly* 60 seconds in one minute.
- There are *approximately* 30 days on average in a month.
- We might **assume** the mass of a typical family car is 1000 kilograms.

It's important to clearly communicate any assumptions made, such as the 'educated guess' above.

Example 1

A typical school year contain 190 teaching days. Estimate how many times a teacher might take a class register during their career, stating any assumptions required.

You may have estimated a different duration for a teacher's career, and a different number of registers taken per day. This is okay, within reason. Your final answer will vary based on your *assumptions* of the different *variables*.

For each Fermi estimation, state clearly any assumptions made.

Example 2

Estimate the total volume of water that a typical person drinks during their childhood.

Example 3

Estimate how much on average a large supermarket might spend on staff wages each week.

Example 4

Estimate how many miles are driven by cars on Scottish roads in a month.

Example 5

Estimate how much of the time during a school musical is taken up by musical songs and performances.

1.2 Modelling Relationships using Graphs

We may wish to explore the *relationships* between two or more variables, such as when considering:

Questions

- How will the **money** in my savings account grow over time if I make regular payments?
- What might a car's **braking distance** by when travelling at different speeds?
- How might the distance travelled in a taxi be related to the **fare**?

We should aim to identify where possible for any relationship:

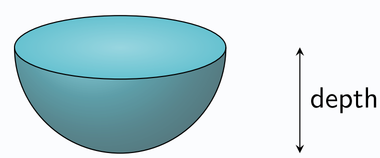
- The independent variable - commonly either *time* or one which we control.
- The **depdendent** variable - which changes in **response** to the other.

Creating graph of a possible model is a good starting point in understanding any relationship. The *independent* variable should be placed along the (horizontal) x-axis and the *response* variable placed on the (vertical) y-axis.

You may have to *sketch* a possible graph of a relationship, or *select from a choice of graphs*, giving an explanation.

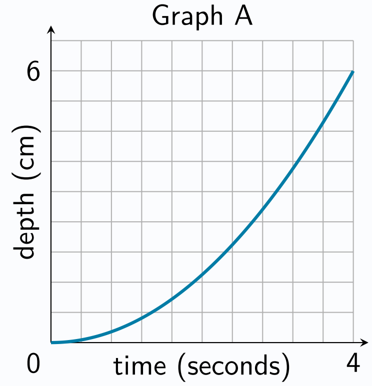
Example

Problem:
Bob pours water into an empty cup at a constant rate.
The interior of the cup is in the shape of a hemisphere, as shown below.

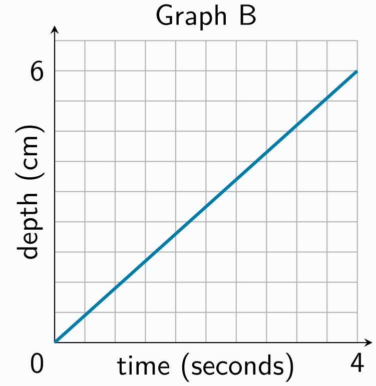


State which of graphs A, B or C below might be best suited to model the relationship between the time since Bob began pouring the water and the depth of water in the cup, explaining your answer.

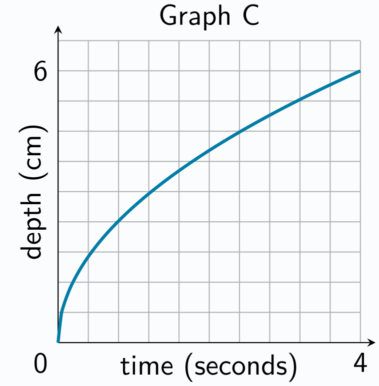
Graph A



Graph B



Graph C

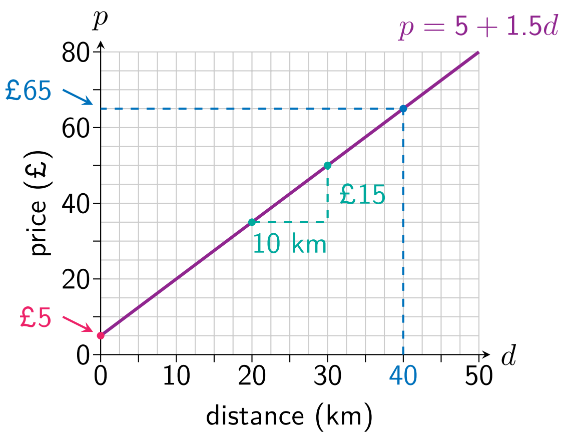


Solution:
Graph C, since the depth will increase more quickly at the start, then slow down towards the end.

1.3 Linear Models

Some types of relationship are commonly encountered, and the rest of this chapter will study these in more detail. One of the simplest relationships between two variables is a *linear relationship*, which on a graph is a **straight line**. A linear relationship is defined by a **constant rate of change**, which can be calculated as the *gradient* of the graph. This tells us the amount of change in the response variable for each positive unit change in the independent variable.

For example, consider the linear relationship between the price of a taxi ride (£p) and the distance travelled (d km).



The **rate of change** is $\frac{15}{10} = £1.50$ per km, so each extra kilometre increases the price by £1.50.

The graph's **starting value**, for a distance of 0 kilometres is £5. (The taxi company's 'base charge').

Note that both the *starting value* and the *rate of change* can be observed from the relationship's **equation**:

$$\boxed{\textbf{price} = 5 + 1.5 \times \textbf{distance}}$$

For any equation of the form $y = a + bx$ describing a linear relationship between x and y, the value of a tells us the *starting value* of y (when $x = 0$), and the value of b tells us how much y changes for each 'step' in the x direction.

The *dashed line* shows how the graph can be used to **predict** a price of £65 for a distance of 40 kilometres.

The equation can also be used:

$$\text{price} = 5 + 1.5 \times 40 = 65$$

Example

Problem:
A candle with a height of 20 centimetres is lit. It loses 4 centimetres of height per hour as it burns.

(a) Write down an equation connecting the candle's height and its burn time.

(b) State the units used to measure the rate of change.

(c) State which is the independent variable and which is the response variable.

(d) Use the equation to calculate the height of the candle after 90 **minutes** spent burning.

(e) Draw a graph to show the relationship.

Solution:

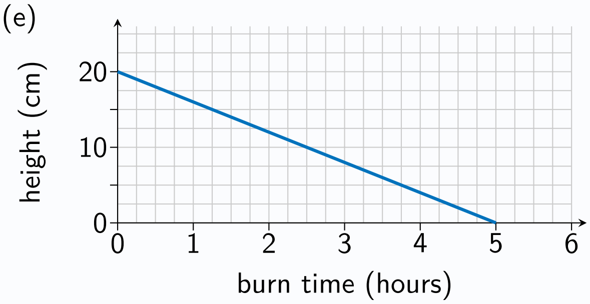
(a) height $= 20 - 4 \times$ time

(b) Centimetres per hour.

(c) The independent variable is time.
The response variable is height.

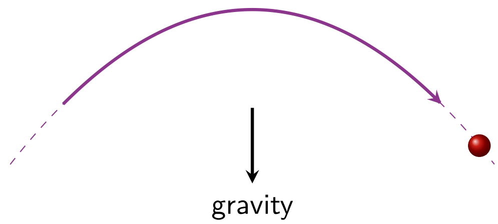
(d) height $= 20 - 4 \times 1.5 = 14$ cm

(e)



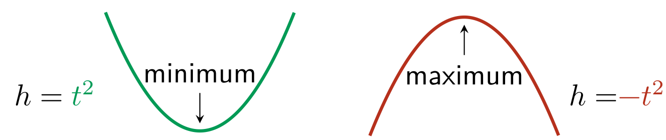
1.4 Quadratic Model

Quadratic relationships occur in a wide range of situations, from the shape of satellite dishes to the design of certain kinds of bridges. One of the most common applications of a quadratic equation is when considering any object acting under gravity, such as a ball thrown through the air.



The **shape** of the graph produced is called a **parabola**.

The 'flipped' version is also a parabola, and the squared term's *sign* tells us if it is a **positive** or **negative** parabola.



The typical feature of a parabola is that it either *falls then rises* or *rises then falls* **symmetrically**, with either a highest point or a lowest point in the middle. Studying *quadratic behaviour* in detail forms parts of the National 5 Mathematics and Higher Mathematics courses. Key skills required for the Higher Applications course include:

- **Recognising** a quadratic relationship through inspecting a graph.
- Using an **equation** or **graph** to find the 'output' (dependent variable) for a given 'input' (independent variable).
- Recognising and using the **symmetrical** property of a quadratic relationship.

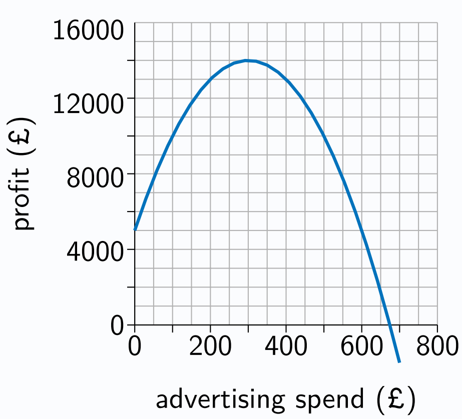
Example 1

Problem:
The height (h metres) of a ball thrown up at a speed of 15 metres per second from 2 metres above the ground t seconds after it is thrown is given by the equation below. Calculate the height after **three seconds**.

$$h = -4.9t^2 + 15t + 2$$

Example 2

The graph below shows the amount of profit a company predicts that they will make for different amounts of money spent on advertising.



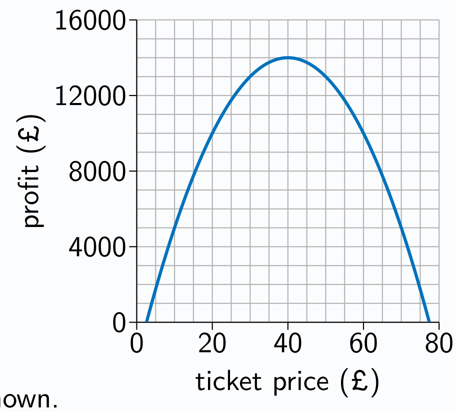
State the type of relationship shown.

Write down the amount of advertising spend which leads to the highest profit, and how much profit the company would expect to make.

Explain what the graph suggests would happen if the company spent more than around £680 on advertising.

Example 3

The graph below shows the profit an events company predicts it will make for different concert ticket prices.



(a) State the type of relationship shown.

(b) Write down the ticket price which leads to the highest profit, and state the highest profit the company would expect to make.

(c) Explain what the model suggests would happen if the company charged more than £80 for each ticket.

Example 4

The height (h metres) of a basketball thrown into the air after t seconds is given by the equation below. Calculate the height after 4 seconds.

$$h = -4t^2 + 15t + 2$$

Example 5

A jet of water is projected upwards from a fountain. The water is at the same height after 3 seconds and after 9 seconds. Its height, h metres, is modelled, where t is the time in seconds after the water is projected, by:

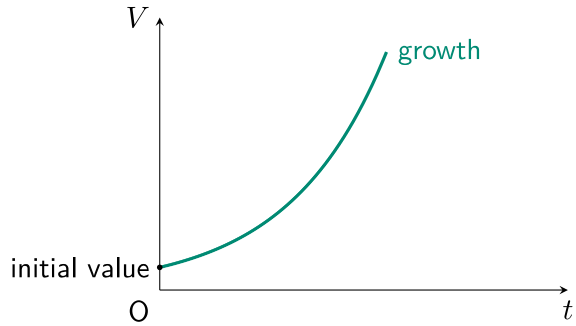
$$h = -2t^2 + 24t + 4.$$

Determine the greatest height reached by the water.

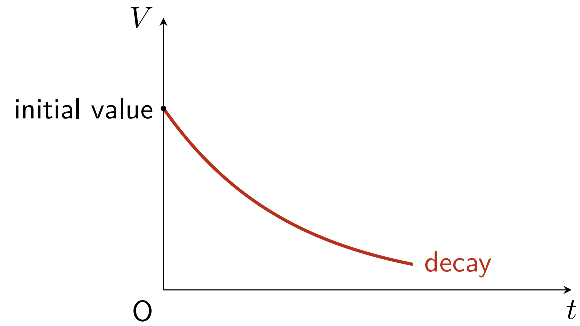
1.5 Exponential Model

The behaviour of many things in the real-world can be appropriately described using an **exponential growth model** or **exponential decay model**. In an exponential model, the independent variable appears as a *power*.

Equations such as $V = 4 \times 1.07^t$ or $V = 4e^{0.068t}$ describe the **exponential growth** of V from some initial value as t increases.



Equations such as $V = 4 \times 0.93^t$ or $V = 4e^{-0.094t}$ describe the **exponential decay** of V from some initial value as t increases.



Exponential growth models describe a dependent variable that **grows faster and faster** as the independent variable increases. For example:

t	0	1	2	3	4	5	6	7
$V = 0.25 \times 2^t$	0.25	0.5	1	2	4	8	16	32

Exponential decay models describe a dependent variable that **decays slower and slower** as the independent variable increases.

t	0	1	2	3	4	5	6	7
$V = 32 \times 0.5^t$	32	16	8	4	2	1	0.5	0.25

A property of a basic exponential relationship is that the dependent variable is multiplied by the same factor over equal intervals of the independent variable. Equivalently, the percentage growth or percentage decay is constant.

A property of basic exponential relationships is that the dependent variable is **multiplied** by the same factor over equal intervals of the independent variable. Equivalent percentage growth or percentage decay is **constant**.

For the **exponential growth** model:

$$\frac{32}{16} = \frac{16}{8} = \frac{8}{4} = 2 \text{ (100\% increase)}$$

For the **exponential decay** model:

$$\frac{2}{4} = \frac{4}{8} = \frac{8}{16} = 0.5 \text{ (50\% decrease)}$$

Example 1

The size of a population R of rabbits t **years after 2026** can be described using a model with equation $R = 400 \times 1.25^t$. Determine the size of the population in 2026.

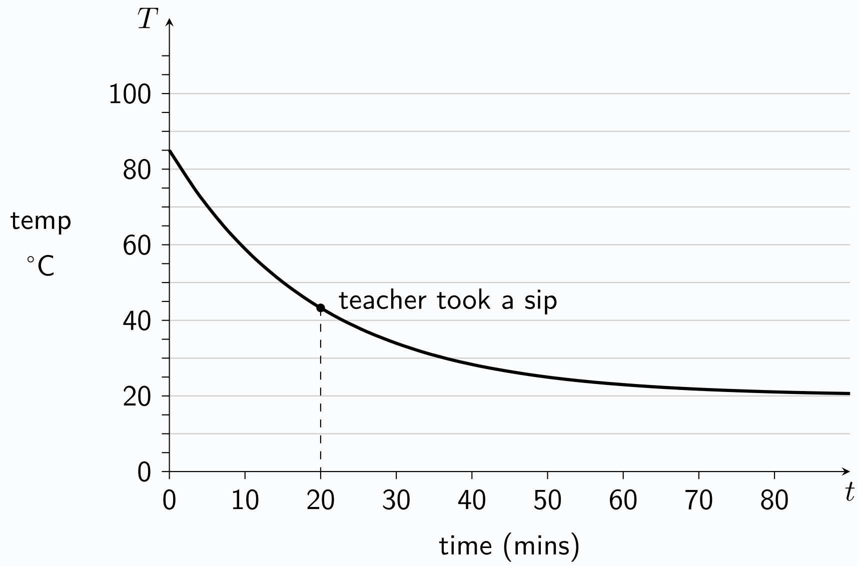
Determine an estimate for the size of the population in 2028.

State what kind of model is being used to describe the size of the population of rabbits over time.

Explain what the number 1.25 represents in the context of the model.

Example 2

A graph is produced showing the temperature T degrees Celsius of the coffee in a mug on a teacher's desk t minutes from the beginning of a lesson.



State the independent and dependent variables.

It is determined that the equation $T = 20 + 65e^{-0.0513t}$ can be used to describe the cooling of the coffee. Calculate the temperature of the coffee at the start of the lesson.

Suggest the likely temperature of the classroom, giving a reason for your answer.

The teacher took a sip of the coffee during the lesson. Calculate the temperature of the coffee.

Example 3

A large pond initially has a small patch of algae on its surface, covering one square foot. Over the next month, the algae spreads across the pond, and an exponential model can be used to describe this.
Suggest a suitable independent variable and a suitable dependent variable for the model.

State the type of exponential model which should be used, giving a reason.

Explain why the exponential model may not be suitable in the longer term to describe the algae's spread.